Evaluation of BERT-Based Models on Patient Data from French Social Media

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BACKGROUND

Social media provide valuable insights about health concerns, living conditions, and clinical histories within general populations or groups sharing a given condition [1]. In Europe, a disease is considered to be rare when affecting fewer than 1 in 2,000 people. Social media have the potential to provide a better understanding of these diseases, including signs, symptoms, and impact on daily life. Among rare diseases, **Developmental and Epileptic Encephalopathies (DEEs) are a group of rare and severe epilepsies**, associated with frequent seizures of different types, and significant developmental delay, regression or plateau [2].

OBJECTIVE

Our aim was to assess the performance of **three transformer models**, namely CamemBERT [4], DrBERT [6] and CamemBERT-bio [5], on a **Named Entity Recognition (NER) task**, in which the entities of interest are phenotypes (all types of clinical signs and conditions present in messages posted on social media) and treatments, using **social media messages from both patients and caregivers discussing health-related topics**. Specifically, the dataset contains testimonies about rare DEEs and messages about common diseases.

Deep learning methods have received attention in recent years for social media analysis and a recent review showed that neural networks perform better than traditional machine learning methods [3]. Large language models exemplified by ChatGPT are undoubtedly powerful tools to discover group themes within text, but **their performance needs to be evaluated in medical domains as specific as rare diseases and on specific types of data such as social media testimonies**. CamemBERT [4], CamemBERT-bio [5] and DrBERT [6] are three language models based on transformers, developed specifically for the French language. CamemBERT-bio, based on CamemBERT, was specifically trained on a French public biomedical dataset [5]; DrBERT is a RoBERTa trained on an open-source corpus of French medical crawled textual data called NACHOS [6].

METHODS

Data and annotation process

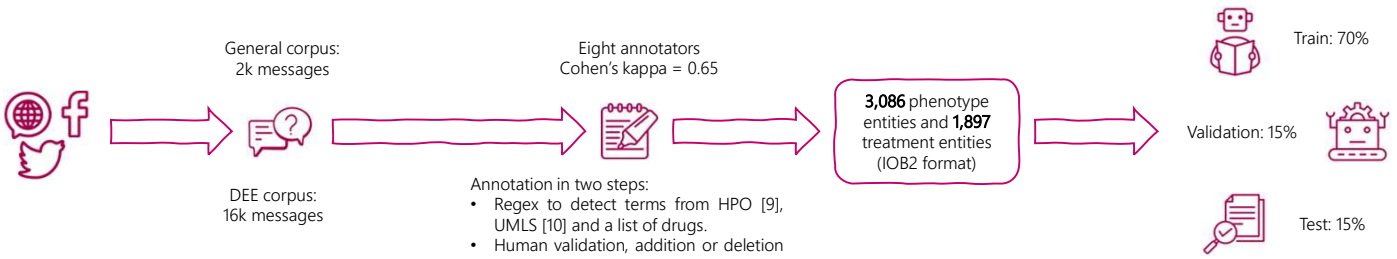


Figure 1 – data and annotation process

Training

All three models, namely CamemBERT, DrBERT, and CamemBERT-bio were trained using the same architecture. They were trained to **optimize the F1-score throughout 10 epochs**. The evaluation strategy involved calculating the model's performance after each epoch. The best-performing model, determined by the **best F1 on the validation dataset**, was kept at the end of the training process. The learning rate was set to 1e-4, controlling the rate at which the model's parameters were updated during training. Both the training and evaluation batches consisted of 8 samples per device. We performed 2020 steps. A weight decay of 0.01 was applied to prevent overfitting during training.

Evaluation

We averaged the results over 10 runs as follows. For each model and each run, the best model on the validation set was used to calculate the different metrics on the test set. The predictions made on the test set were used to calculate the scores, which were then averaged and presented. Scores were calculated using the **seqeval tool on strict mode** [12], the same framework used in [5].

In order to compare the models on our NER task, we calculated **precision**, **recall**, and **F1 score** for each entity as well as the macro, micro and weighted averages. Additionally, we compared our micro-average precision, recall, and F1s with those reported in [5] for other NER tasks, namely EMEA - containing drug leaflets - and MEDLINE - containing scientific article titles [13] which were manually annotated following ten semantic groups from UMLS [10].

RESULTS

		camembert	camembert-bio	drbert
phenotype	f1-score	0.55	**0.56**	0.49
	precision	**0.62**	0.58	0.58
	recall	0.50	**0.53**	0.43
treatment	f1-score	0.64	**0.65**	0.64
	precision	0.58	0.56	**0.62**
	recall	0.72	**0.78**	0.65
micro-average	f1-score	0.59	**0.60**	0.55
	precision	**0.60**	0.57	**0.60**
	recall	0.58	**0.62**	0.51
macro-average	f1-score	0.60	**0.61**	0.57
	precision	**0.60**	0.57	**0.60**
	recall	0.61	**0.66**	0.54
weighted-average	f1-score	0.58	**0.59**	0.54
	precision	**0.61**	0.57	0.59
	recall	0.58	**0.62**	0.51

Table 1 - Performance comparison -using seqeval [12] in strict mode- of CamemBERT-base, CamemBERT-bio-base and DrBERT-7GB on NER task on social media testimonies.

	EMEA			MEDLINE			Social Media Testimonies		
	weighted	macro	micro	weighted	macro	micro	weighted	macro	micro
drbert	0.87	0.35	-	0.76	**0.15**	-	0.54	0.57	0.55
camembert-bio	**0.90**	0.36	0.77	**0.78**	**0.15**	0.68	**0.59**	**0.61**	**0.60**
camembert	0.88	**0.47**	-	0.76	0.12	-	0.58	0.60	0.59

Table 2 - Comparison of F1-scores -using seqeval [12] in strict mode- of CamemBERT-base, CamemBERT-bio-base and DrBERT-7GB on three different NER tasks: EMEA, MEDLINE [5] and our NER task.

DISCUSSION

The main result of this study is the similar performance achieved by a general model - CamemBERT - and a domain-specific one - CamemBERT-bio - in identifying phenotypes in social media testimonies. These results suggest that both models effectively leverage the language representations learned from French general and health-related data. DrBERT's performance was slightly lower on phenotypes but better on treatments. Overall, **the weighted-average F1-score was the same for CamemBERT and CamemBERT-bio and slightly lower for DrBERT**.

The comparison of F1-scores across three different NER tasks - EMEA, MEDLINE, and our social media testimonies task - indicates that **the weighted F1-scores for both EMEA and MEDLINE tasks were notably higher** (.87-.90 and .76-.78, respectively) compared to our social media testimonies task (.54-.59). This discrepancy suggests that the models perform better when applied to more structured and domain-specific datasets, such as those found in regulatory or biomedical literature contexts.

CONCLUSIONS

This preliminary work revealed that on social media testimonies, **CamemBERT and CamemBERT-bio have almost the same performance, slightly higher than DrBERT**. It also highlighted that **the performance on social media testimonies is still lower than on structured and more traditional health datasets** like MEDLINE. Models capable of accurately extracting phenotypes and treatment entities from social media messages could help in better understanding the challenges faced by patients and their caregivers, especially in the rare disease and DEEs field.

Several limitations should nevertheless be addressed. The evaluation primarily focused on NER performance, neglecting other important tasks. Future research is required to provide a more comprehensive assessment of model capabilities. Additionally, the generalizability of our findings may be limited by the specific dataset and evaluation setup used in this study. Further research involving larger and more diverse datasets is needed to validate and extend our findings.

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